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Unit 5 : Session 5 : SLO 1 | Algebraic Simplification Using SymPy

I. Questions & Answers

1. Simplify $(x^2 - 1)/(x - 1)$ → `simplify((x**2 - 1)/(x - 1))` → $x + 1$ (the factor $x - 1$ cancels; valid for $x \neq 1$).
2. What `simplify()` does → applies a battery of heuristics (algebraic, trigonometric, logarithmic rules) to rewrite an expression in a simpler, more compact equivalent form.
3. Expand $(x+1)(x-1)$ → `expand((x + 1)*(x - 1))` → $x^2 - 1$ (difference of squares).

II. Code

- `from sympy import symbols, simplify, expand`
- `x = symbols('x')`
- `print(simplify((x**2 - 1)/(x - 1)))` # $x + 1$
- `print(expand((x + 1)*(x - 1)))` # $x^2 - 1$